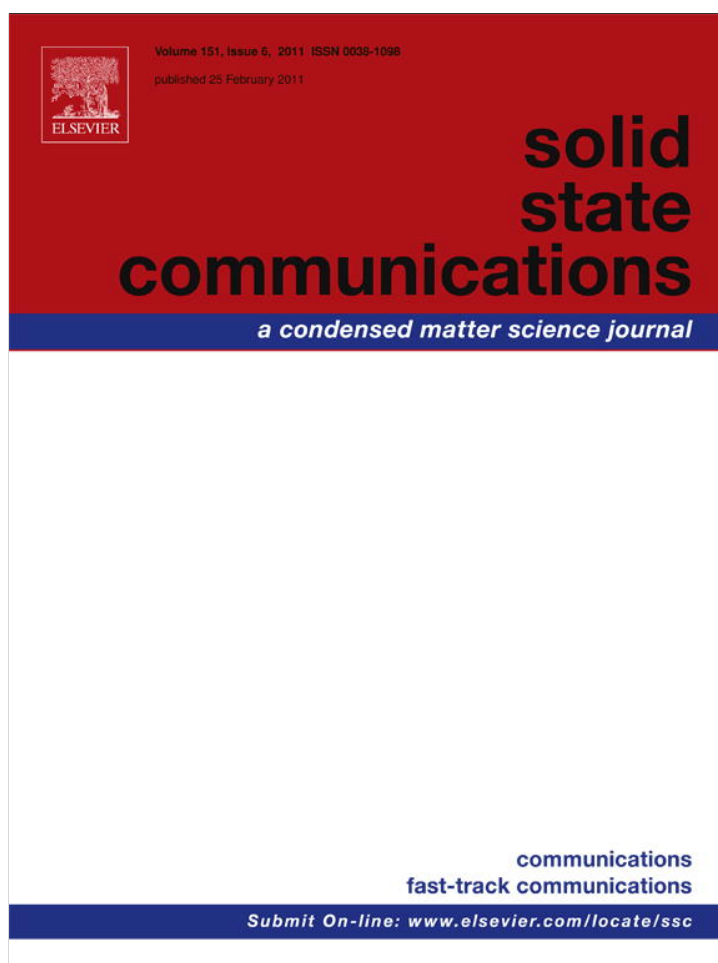




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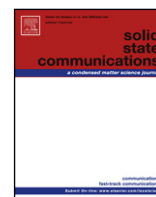
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journal homepage: www.elsevier.com/locate/sscAn atomistic approach to the dielectric modes of BaTiO₃ and SrTiO₃

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ABSTRACT

Recently, a clear long-sought Debye mode (*D*) in barium titanate (BT) was identified [J. Hlinka, T. Ostapchuk, D. Nuzhnyy, J. Petzelt, P. Kuzel, C. Kadlec, P. Vanek, I. Ponomareva, L. Bellaiche, Phys. Rev. Lett. 101 (2008) 167402] and this mode was shown to originate from the polar coordinate that also generates the so-called Slater mode (*S*). The inter-relations between the *D* mode and the normal A1-type phonon modes were studied by those authors using a four-mode phenomenological model. The present work is to offer an atomistic support of their work and to better illustrate the nature of and the couplings between these modes. In addition, we extend the as-obtained insights to the investigation of the nature of the low frequency Raman peaks that were found in ¹⁸O-substituted strontium titanate (ST) many years ago [M. Itoh, R. Wang, Y. Inaguma, T. Yamaguchi, Y.-J. Shan, T. Nakamura, Phys. Rev. Lett. 82 (1999) 3540]. Thus, our work provides important information on the connections between the lattice dynamics of BT and ST.

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1. Introduction

Soft modes in the ferroelectric transition, particularly in perovskite oxides, has been extensively examined [1–11]. A recent work on barium titanate (BT) has shown convincingly that this mode is of relaxation type and can be well fitted by Debye formula [12]. The nature of the relaxational *D* mode and the normal A1-type phonon modes, i.e., the so-called Last (*L*), Slater (*S*) and Axe (*A*) modes [13,14], was studied by a phenomenological model [12]. It was claimed that the *D* and *S* modes should originate from the same polar coordinate [12]. The *D* mode is supposed to be associated with the large amplitude inter-well hops of this coordinate while the *S* mode is associated with the small amplitude intra-well vibrations [12]. Since the motions of Ti atoms are the main source of the dielectric response [1], this coordinate describes basically the motions of Ti atoms, as was theoretically suggested [15,16] and experimentally verified [17,18]. Hence, in a simplified picture, the *L* and *A* modes should largely come from the vibrations of O atoms.

Differing from BT, which is a typical ferroelectric, strontium titanate (ST) is usually quoted as a quantum paraelectric [19,20]. It displays a quite different *S* mode from that observed in BT, though the *L* and *A* modes are similar [21,22]. The Debye-like mode is absent in ST [21,22]. Considering the similarity in crystal structure between ST and BT, the origin of the absence of the *D* mode needs to be answered. Does the main difference in dielectric modes between ST and BT stem from the difference in the motions of Ti atoms? Moreover, the doping of ¹⁸O into ST turns it from a paraelectric to a ferroelectric [23]. It was observed that, at extremely

low temperatures there are two pronounced peaks located below 20 cm^{−1} in the Raman spectrum of the ¹⁸O-substituted ST (ST18) [23]. However, these peaks were barely discernible in ST. What is the nature of these peaks?

The purpose of the present work is two-fold. First, we want to construct an amenable atomistic model to better illustrate the nature of and the inter-relations between these dielectric modes. Second, we want to show that the aforementioned Raman peaks in ST18 can be reproduced by this same model. According to our model, these peaks stem from the strong coupling between *T* (a tunneling mode) and *L* modes.

2. Theory

In this model, we only consider the motions of the Ti- and O-subsystems, while the Ba (or Sr)-subsystem can be treated frozen as judged from the partial densities of phonon modes [24]. For simplicity we confine ourselves to the dielectric response under an external field along the *z*-direction. To construct the model, for simplicity, we first assume that each Ti atom is subjected to a double-well potential $V_B = bz_i^4 - az_i^2$, where *b* and *a* are positive constants and *z_i* is the *z*-component of the displacement vector of the *i*-th Ti atom. Such a potential is well established both theoretically [9,10,16,25] and experimentally [17,18].

In tetragonal phase, there are two non-equivalent O sites, the transverse ones O_{xy} located in the *x*-*y* plane and the longitudinal one O_z sitting along the *z*-axis. The potential is assumed to be double-well for the planar O atoms but single-well for O_z. This treatment is based on the O anisotropic shell polarizability and covalent bonding [4,26]. The corresponding vibration frequencies are denoted by ω_{*t*} and ω_{*l*} for O_z and O_{xy}, respectively. Now given

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that the Ti–O σ -type bond favors motion along the bond whereas it prevents motion perpendicular to it [26], one has $\omega_t > \omega_l$. As one moves from BT to ST, all bonds become stronger and thus transverse motions become more difficult while longitudinal motions become easier and, as a result, ω_t increases while ω_l decreases. Comparing this tendency with observation [21], one naturally associates the L mode with ω_l and the A mode with ω_t . Furthermore, the D and S modes can be linked to the motions of Ti ions, as conjectured in Ref. [12]. Finally, for the O-subsystem, the potential may be written as $V_O = \frac{m_{t,l}}{2} \omega_{t,l}^2 z_{t,l}^2$, where t and l refers to transverse and longitudinal motions, respectively, and $z_t = \frac{1}{2}(z_{Ox} + z_{Oy})$, $z_l = z_{Oz}$, $m_t = 2m_o$, as well as $m_l = m_o$, with m_o being the mass of O core. One can show that the variable $\bar{z} = \frac{z_{Ox} - z_{Oy}}{2}$ is neither coupled to z_α nor to Ti atoms and nor to IR light for zone center mode. ω_l can be determined by the self-consistent phonon approximation (SPA) [27].

An essential ingredient of the model is the coupling between the O- and Ti-subsystems, which will be chosen in a bilinear form of the coordinates of either subsystem, that is, $V_{BO} = \beta_{ij}^t z_i z_j^t + \beta_{ij}^l z_i z_j^l$. For simplicity, only the interaction between nearest neighbors is taken into consideration, that is, $\beta_{i,j}^t$ ($\beta_{i,j}^l$) is zero unless i and j label adjacent sites, in which case the coefficient will be $2\beta_t$ (β_l). This coupling has two ubiquitous origins. One is the direct dipolar interaction and the other comes from O polarization. The latter takes place between an O ion and a Ti one if they are interacting with a common O electron cloud, which acts as a glue to bind them. Furthermore, the former is repulsive and tends to alienate them. Putting all the contributions together, the full model can be written as $H = KEB + KEO + V_B + V_O + V_{BO}$, where KEB and KEO stand for the kinetic energies of Ti- and O-subsystems, respectively.

As a primitive treatment, the adiabatic approximation [9] can be invoked to decouple the subsystems, which results in $H_B = KEB + V_B - K_{ij} z_i z_j$ and $H_O = KEO + \frac{m_{t,l}}{2} \omega_{t,l}^2 U_{t,l}^2$, which govern the motions of Ti- and O-subsystem, respectively. Here $U_{t,l}^i = z_{t,l}^i + d_{t,l}^i$, with $d_{t,l}^i = \frac{2\beta_{t,l}(z_i)}{m\omega_{t,l}^2}$ and where K_{ij} are nonzero only when the i -th and j -th Ti cores are nearest neighbors, in which case it becomes $K_l = \frac{2\beta_l^2}{m\omega_l^2}$ for

Ti cores lying along the z -direction and $K_t = \frac{2\beta_t^2}{m\omega_t^2}$ for Ti cores lying along either the x - or y -directions. The physics of H_B are well known and two points should be mentioned: (1) due to the competition between quantum fluctuations (quantified by $\frac{b}{4}(\frac{4}{a})^{3/2} \frac{\hbar^2}{m\eta}$) and ordering (quantified by $(2K_l + 4K_t)/a$), a quantum critical point exists separating the ferroelectric (FE) phase from the paraelectric one [28]; (2) two modes can be seen, one around zero frequency, with the other at a much higher frequency, whose intensities and frequencies change with temperature [29,30]. Hence, compatible with Ref. [4], four dielectric modes can be obtained in BT and the association of S and D modes with the Ti subsystem is evident. In cubic BT, a recent study [31] revealed two modes below 150 cm^{-1} , one tending to zero and the other saturating at 60 cm^{-1} as the temperature goes down to T_C . The former may be assigned to the D mode while the latter be assigned to the S mode.

The above adiabatic treatment neglected dynamic O–Ti interactions, which is not realistic. Hlinka et al. clearly demonstrated the inter-mode coupling [12] in their model. However, they considered only the coupling of the S mode to the other ones. Considering that both D and S modes originate from the motion of Ti ions, it is inappropriate to ignore the coupling of the D mode to the others.

To proceed, let us put $z_i = l_0 \sigma_i + \eta_i$, where $l_0 = \sqrt{\frac{a}{2b}}$ stands for the potential well position of V_B . In general, both σ_i and η_i are dynamic variables contributing to the kinetic energy KEB . The main advantage of this expression is to represent inter- and intra-well

motions by different variables, better reflecting the origin of various response modes. As will be shown below, σ_i describes inter-well hopping and η_i pertains to intra-well oscillations. Inserting them into H , we come to the following equations

$$H' \approx H_{osci} + H_{softspin} + H_{o-s} \quad (1)$$

$$H_{osci} = KEO + V_O + \frac{m_{Ti}}{2} \omega_s^2 \eta_i^2 - \frac{\theta \hbar^2}{2m_{Ti}} \frac{\partial^2}{\partial \eta_i^2} \quad (2)$$

$$H_{softspin} = -\frac{(1-\theta) \hbar^2 b}{am_{Ti}} \frac{\partial^2}{\partial \sigma_i^2} + \frac{a^2}{4b^2} (\sigma_i^4 - 2\sigma_i^2) \quad (3)$$

$$H_{o-s} = C \sigma_i \eta_i + \beta_{ij}^{t,l} (l_0 \sigma_i + \eta_i) z_j^{t,l} \quad (4)$$

Here $C = 4bl_0 \langle \eta_i^2 \rangle$ and $\frac{m_{Ti}}{2} \omega_s^2 = \frac{2a}{b} + b \langle \eta_i^2 \rangle$ take on the usual mean-field form, where we have used $\frac{\partial^2}{\partial z_i^2} = \frac{\partial^2}{\partial \eta_i^2} = \frac{2b}{a} \frac{\partial^2}{\partial \sigma_i^2}$. The parameter θ is introduced to control the dynamics. We will only be interested in two cases: $\theta = 0$ and $\theta = 1$ [32]. It should be noted that ω_s cannot be simply determined through SPA, because it may be significantly influenced by soft spins. Thus we have arrived at a system of soft spins [33] and oscillators, as indicated by the subscripts of the terms of H' .

3. Results and discussion

If l_0 is large, inter-well tunneling effects will be negligible and FE can be expected below T_C . BT is an example of this situation. At relatively low temperatures, a Ti atom will mainly stay around the bottom of either well and inter-well transitions can only be attained through thermal fluctuations. Only when the kinetic energy suffices to overcome the energy barrier can coherent transitions occur. Therefore we introduce $T^* \geq T_C$ to mark the temperature above which coherent transitions take place. Above T^* , the spins are indeed soft and $\langle \sigma_i^2 \rangle$ can have significant fluctuations about 1. Thus, it will be convenient to set $\theta = 0$ and $\eta_i = 0$ so that σ_i dominates the displacement of Ti ions, resulting in three modes. Below T^* , the spins become stiff and the $\langle \sigma_i^2 \rangle$ have little fluctuation, which suggests $\theta = 1$. In this case the spin dynamics are absolutely incoherent, of relaxational type, and four modes should be anticipated. Therefore, T^* marks the temperature at which a crossover from four-mode to three-mode response might take place. Detailed analysis along this line is beyond the scope of the present communication. At low temperatures, η_i should be very small in comparison with l_0 , which means one can neglect the coupling between η_i and the other variables. Following Ref. [9], we obtained the dielectric response to a uniform external electric field along the z -axis. Below T^* , it can be expressed as $\varepsilon(\omega) = \varepsilon_\infty + \varepsilon_s + \frac{4\pi}{a_0} Z_B^2 e^2 Q^T [D'(\omega) - i\omega D''(\omega)]^{-1} Q$, where $Q^T = [v_B, v_l, 2v_t]$ denotes the transpose of Q (the effective charge vector in units of $Z_B e$), a_0 the lattice constant, e the elementary charge, and Z_B the valence of Ti ion ($Z_B = 4$). Additionally, we have $\varepsilon_s = S/(\omega_s^2 - \omega^2 - i\omega\Gamma_s)$ and $D''(\omega) = \text{diag} \left[\frac{\tau k_B T}{l_0^2}, m_l \Gamma_l, m_t \Gamma_t \right]$ as well as

$$D'(\omega) = \begin{pmatrix} k_B T / l_0^2 (1 - \sigma^2) & 2\beta_l & 4\beta_t \\ 2\beta_l & m_l (\omega_l^2 - \omega^2) & 0 \\ 4\beta_t & 0 & m_t (\omega_t^2 - \omega^2) \end{pmatrix} \quad (5)$$

where the dissipation coefficients $\Gamma_{s,t,l}$ have been included, τ represents the relaxation time of spins and S denotes the intensity of the S mode. Further, σ is the mean value of spins, which vanishes above T_C and goes as $\sqrt{\frac{T_0 - T}{T_0/3}}$ just below T_C .

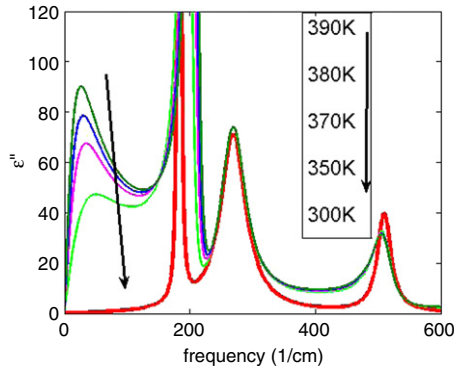


Fig. 1. The imaginary part of the dielectric function of BT at various temperatures below T_C .

Here T_0 is a little bit larger than T_C , reflecting the first order transition. One can show that above T_C the static dielectric function $\epsilon(0)$ assumes the Curie form, with $T_C = \frac{4l_0^2}{k_B} \left(\frac{2\beta_t^2}{m_0\omega_t^2} + \frac{\beta_l^2}{m_0\omega_l^2} \right)$ and the Curie constant is given by $C = \frac{4\pi Z_B^2 e^2 l_0^2}{k_B a_0^3} \left[v_B^2 + \frac{v_t^2 k_B T_C / l_0^2 - 4v_l v_B \beta_l}{m_0\omega_l^2} + 2 \frac{v_t^2 k_B T_C / l_0^2 - 4v_l v_B \beta_l}{m_0\omega_t^2} - \frac{8(v_l \beta_l - v_t \beta_t)^2}{m_0^2 \omega_l^2 \omega_t^2} \right]$. Using proper parameters [26], we can get $T_C \approx 410$ K and $C \approx 10^5$ K for BT. The corresponding calculated frequency dependence of dielectric function is displayed in Fig. 1, which agrees reasonably well with experimental results [13]. Interestingly, we found $T_C \propto d_{Ti}^2$, in agreement with the empirical AKJ relation [34]. It can be seen from Fig. 1 that, as the temperature approaches T_C , the lowest frequency mode sharpens rapidly and disperses to lower frequencies, indicating a relaxation-type mode. Since we did not take into account the temperature dependence of ω_s and its coupling to other modes, the peak around 270 cm^{-1} shows no discernible shifts and its weight remains unchanged. The width of the peak near 180 cm^{-1} (L mode) is quite sensitive to temperature and so is the shape of the peak at the highest frequency (A mode). This indicates a temperature dependent effective mode-coupling. The present result resembles that obtained via a phenomenological model [12].

If l_0 is very small, quantum tunneling will be strong [28], which suggests $\theta = 0$ and $\eta_i \equiv 0$. ST is typical of this case. At high temperatures, where quantum fluctuations are overwhelmed by thermal fluctuations, from SPA one gets, $\sigma_i^4 - 2\sigma_i^2 \approx \sigma_i^2 ((\sigma_i^2) - 2)$ and $H_{softspin}$ reduces to the Hamiltonian of harmonic oscillators. Thus, the dielectric response deduced from H' consists of three resonances (L , S , and A modes), in accord with experiments [21,22]. It should be noted that the S mode for ST is at about 80 cm^{-1} , close to that of 60 cm^{-1} for BT [31]. At low temperatures, where quantum effects prevail [11], we may approximate the soft spins as Ising ones and the kinetic energy may be affected by a spin-flip term, $\Omega_0 \sigma_i^x$, which gives rise to the T mode, which is a tunneling mode of frequency $\Omega = 2\Omega_0/\hbar$ [35]. With the means prescribed in Ref. [36] and the parameters in [27], we are able to obtain the low temperature Raman spectrum for ST, as displayed in Fig. 2. The agreement with Ref. [23] is remarkable: below 20 cm^{-1} , two modes are found, which originate from the T mode at Ω and the L mode. Incidentally, we also found that $\text{SrTi}^{18}\text{O}_3$ is ferroelectric, with T_C of around 5 K, whereas ST is not.

Finally, we notice that ST can be taken as a highly anisotropic Ising system in a transverse field of strength Ω_0 . This is evident if we compute K_t/K_l , which increases rapidly to a huge value at low temperatures, primarily due to the softening of ω_l . A corollary of this anisotropy is the formation of one-dimensional texture resulting from strong chain-like spin-spin correlations at a temperature far above T_C [37]. MD simulations found support for this scenario [38] in BT and first-principles calculations also advocated this [39].

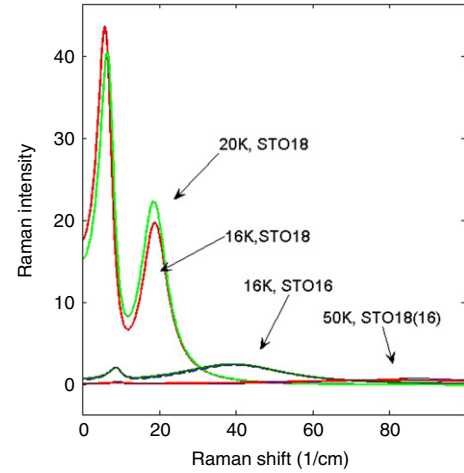


Fig. 2. Raman spectrum of ST at low temperatures.

4. Conclusions

Our model provides an atomistic view for the phenomenological description of the four dielectric modes observed in BT. Significant couplings exist between the D mode and the L , S , and A modes, while the couplings between the S mode and the others are negligible at very low temperatures. A crossover from the high temperature 4-mode to the low temperature 3-mode behavior is predicted in BT. Using the same model, we have reproduced the Raman peaks that were observed in ^{18}O -substituted ST, but are absent in ST. Moreover, there is no similar crossover in this compound as the temperature is varied. Finally, we conclude that ST can be modeled by a strongly anisotropic Ising Hamiltonian in a transverse field.

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